



Tutorial exercises (1)

1. EXERCICE

1. Describe the following concepts:
 - Algorithm , Program ,Data structure ,constant ,variables
2. Explain the resolution cycle of a computer problem.
3. Give the general form of an algorithm
4. Descript the steps to be taken to complete the following tasks :
 - Swap the contents of two cups,one containing milk, the other water.
 - Get the sum of (4) four numbers.

2. EXERCICE

A salesman has an 8-litre can of oil. He wants to divide the can into two equal parts (4 litres for each part) using two jars of 5 and 3 litres respectively.

- Analyse this problem and express the solution as an ordered sequence of steps.

3. EXERCICE

A man dies leaving 17 camels (or other animals) to his three sons, to be divided in the following proportions: the eldest son should inherit $\frac{1}{2}$ of the man's property, the middle son should inherit $\frac{1}{3}$, and the youngest son should inherit $\frac{1}{9}$.

- How should they divide the camels, noting that only a whole live camel has value?

4. EXERCICE

There are 3 numbers A, B and C in 3 boxes X, Y and Z. We want to rearrange these numbers so that box X contains the smallest number and box Z the largest.

- Give the steps of the solution.

5. EXERCICE

Find the missing numbers :

- 1,8,27,64, X

- | | | | | | |
|----|----|----|----|----|----|
| 4 | 5 | 6 | 7 | 8 | 9 |
| 61 | 52 | 63 | 94 | 46 | XX |

6. EXERCICE

You have 12 visually identical balls. Eleven of them share the same weight, while one the anomalous ball differs in mass, though you don't know whether it's heavier or lighter than the rest. Using a two-pan balance scale, you can perform up to three weighings. Each weighing reveals whether the pans are balanced or, if unbalanced, which side is heavier.

Your challenge: Identify which ball is the anomaly and determine whether it's heavier or lighter than the normal balls—all within exactly three weighings.

7. EXERCICE

- How $2036 : 4 = 10$?
- Consider the sequence : 1,11,21,1112,3112,...
What is the following element?

8. EXERCICE

For each of the following expressions, determine:

1. The **order of evaluation** (the sequence in which the operations are performed).
2. The **final result** of the expression.
3. The **data type** of the final result.

■ $(-5 * 3) + (5 - 2) - (4 \uparrow 2)$

■ $9 * ((12 - 3) / (1 + 2)) \uparrow (2 + 2)$

■ $-3 * (2 + (5 - 9)) - (2 * 4) \uparrow 2$

■ $(P \text{ and True}) \text{ or } (\text{not } P \text{ and not True})$

■ $(P \text{ or not } P) \text{ and } (\text{True and True}) \text{ and } (5 > 3)$

■ $(3 > 2) \text{ and } (21 = (19 + 2)) \text{ or } (8 < 10)$

9. EXERCICE

Find the boolean values taken in the following algorithm :

```

Algorithm exercise_bool;
Var a,b:integer ;
b1,b2,b3,b4:boolean;
Begin
a ← 10;
b ← 4;
b1 ← (10 > 10)and(5 = 5);
b2 ← (a = 10)Or(b = 5)or(3 = 6);
b3 ← (a > b)and(5 = 5)or(b < a);
b4 ← (False)and((True)or(a > b));
End.

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10. EXERCICE

Given the following Algorithms:

Algorithm A ; Var a,b,c:Integer ; Begin a ← 5; b ← a + 1; c ← a + b; End.	Algorithm B ; Var a,b,c,d:Integer ; Begin Write('Give a and b :'); Read(a,b); c ← a + b; d ← a * b; Write(c,d); End.	Algorithm C ; Var x,y:Integer; Begin Write(Give x and y :); Read(x,y); x ← x * y + 1; z ← z + x; Write('x,z'); End.	Algorithm D ; Var a,b,c:Integer Begin Write(Give a and b :); Read('a,b'); c ← a a ← b; b ← c; Write('a,b'); End.
	a=5 and b=6 a=6 and b=5	x=9 and y=3	a=5 and b=7

- **Algorithm A**

1. trace the algorithm (A) and give the values of variables (a),(b) and (c).

- **Algorithm B**

1. What does this algorithm do?
2. Trace the algorithm for a=5 and b=6.
3. Same question for a=6 and b=5.

- **Algorithm C**

1. Correct the algorithm if there are errors.
2. Roll out the algorithm for x=9 and y=3.

- **Algorithm D**

1. Does the algorithm contain errors?
2. Roll out the algorithm for a=5 and b=7 and deduce what it does.